Puzzle × Story Rich: the white space in Steam's indie tag network

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

1 The decision

Which two-tag concept the **creative director of a hypothetical 5-person Bangkok indie studio** pitches at the greenlight on **Mon 2 Nov 2026**; it fixes the next 18-month project's genre, scope and Steam-page tags. Had no pair passed the white-space test, the pitch would be a fast follower in a proven niche.

2 Why a network

A tag-count spreadsheet ranks popular tags; it cannot see whether two tags sit in *separate* audience neighbourhoods, i.e. whether combining them bridges a structural hole. That needs neighbours and paths. Prior Steam-tag networks (Li & Zhang, 2020) were descriptive; this one drives a build decision.

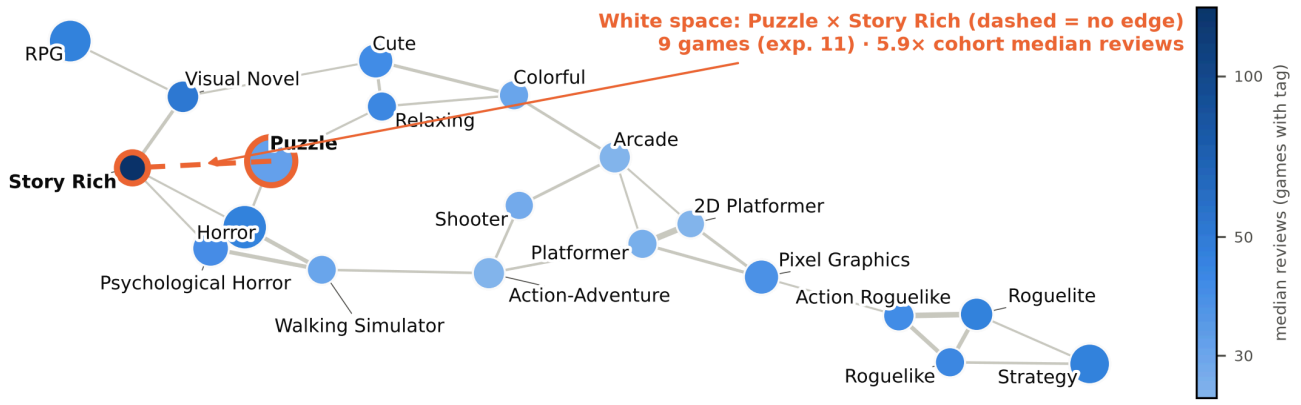
3 Data

Sources	**S1** FronkonGames Steam Games Dataset (Hugging Face): frame + snapshot. **S2** live Steam store, 2026-10-14: tag votes (app page), review totals (appreviews).
Legal	S1 MIT licence. Steam robots.txt allows /app/, /appreviews/; 1 req/1.5 s, honest user-agent. Game-level only, no reviewer data (PDPA n/a).
Volume	Indie, released 1 Jan 2023 – 31 Dec 2024, ≥10 reviews: 9,094 games → random 1,000 (seed 7202); 996 with live tags.
Cross-check	S1 vs S2 top-5 tags: mean Jaccard 0.87; reviews ρ = 0.98. SteamSpy was tested and dropped: it mirrors S1 exactly (Jaccard 1.00).
Bias	Reviews count buyers of existing games, not unmet demand; the ≥10-review floor favours games that got traction; tagging is English-centric.

4 Graph construction

Node, edge	A tag. An edge asserts both tags are among the same game's top-5 tags by votes.
Excluded	Seed "Indie"; tags on >25% of games (Singleplayer, Adventure, Casual, Action, 2D, 3D, Atmospheric, Exploration, Simulation, First-Person); release-model and adult tags.
Weight	**NPMI, not raw count.** By raw count 9 of the top-10 edges touch the 5 commonest tags (top: Action–Adventure, 66 games); by NPMI, 0.
Window	**Top-5 tags, not all 20**: the store shows the top tags. With all 20, 72 games carry the winning pair (mostly as minor tags) and its lift falls from 5.9× to 1.9×.
Pruning	Tag on ≥15 games; NPMI ≥ 0.10 and ≥5 shared games. 391 tags → **98 nodes, 249 edges**.

5 The picture · size = games with tag · colour = median reviews · edge width = NPMI



6 Findings

- **Puzzle is the main broker** (betweenness 0.17, constraint 0.13), linking 14 tags across separate neighbourhoods.
- **White space: Puzzle × Story Rich.** Only 9 games combine them (chance expects 11) and their neighbourhoods barely overlap (0.10), yet their median game has **302 reviews, 5.9× the cohort's 51** (permutation p = 0.021); 22% are top-decile hits vs 10% overall.
- **Crowded vs open.** The top-lift pair, Management × Strategy (13.0×), is 2.4× over-supplied. Ours is #1 at all 8 cutoffs and on S1 tags; lift > 1 in 99/100 bootstraps (90% CI 1.6–39×), but #1 in only 3/100: with n = 9, its *rank* is fragile.

7 Recommended action

- **Mon 2 Nov 2026:** creative director pitches Puzzle × Story Rich at greenlight; plan B is the fast follower Management × Strategy.
- **2 Nov – 11 Dec:** 6-week vertical slice. **Mon 14 Dec:** Steam "Coming Soon" page live with Puzzle and Story Rich as its first two tags.
- **Fri 15 Jan 2027, go/no-go:** ≥1,500 wishlists in 30 days → full production; else plan B and re-run on 2025 releases.

8 What would change my mind

Co-occurrence is not causation, and 9 games is a thin base one breakout could carry. If the wishlist test misses or the pair loses its lead on 2025 releases, I switch to plan B (Management × Strategy).